

Week 3 Assessment — Data Visualization with ggplot2

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Exercise 1 — Reproducing and combining ggplot2 figures

The Two figures

```
nh <- read_csv("Data /cleaned_NHANES.csv")
# Adjust the column names below to match the actual cleaned dataset
# (e.g., age, gender, ethnicity)
```

```
# Figure 1: Histogram of age by gender
p1 <- nh %>%
  ggplot(aes(x = age, fill = gender)) +
  geom_histogram(binwidth = 5, position = "dodge", alpha = 0.7) +
  scale_fill_manual(values = c("salmon", "#008080")) +
  labs(
    title = "Distribution of Age by Gender",
    x = "Age (years)",
    y = "Count"
  ) +
  theme_minimal()
  theme(plot.title = element_text(size = 12))
```

```
## <theme> List of 1
## $ plot.title: <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : num 12
## ..@ hjust       : NULL
## ..@ vjust       : NULL
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : NULL
## ..@ debug       : NULL
## ..@ inherit.blank: logi FALSE
## @ complete: logi FALSE
## @ validate: logi TRUE
```

p1

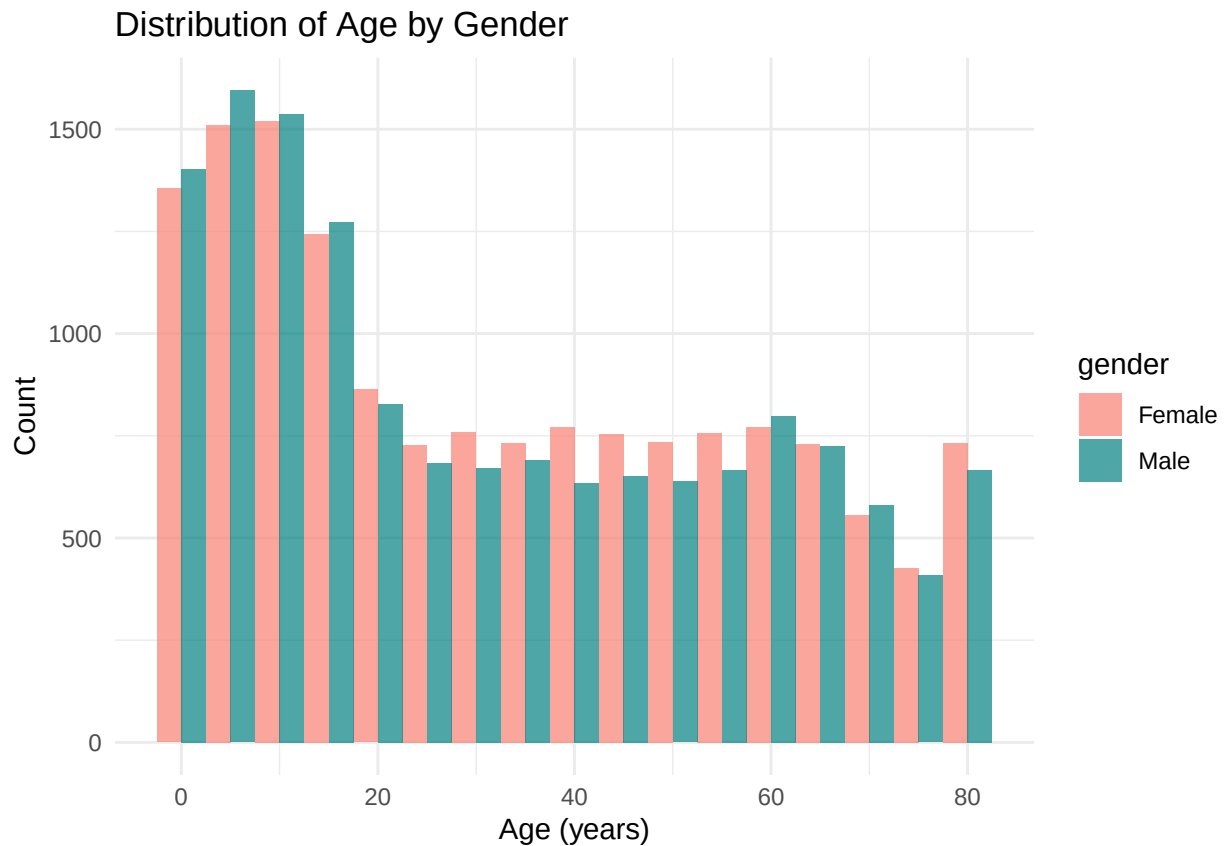


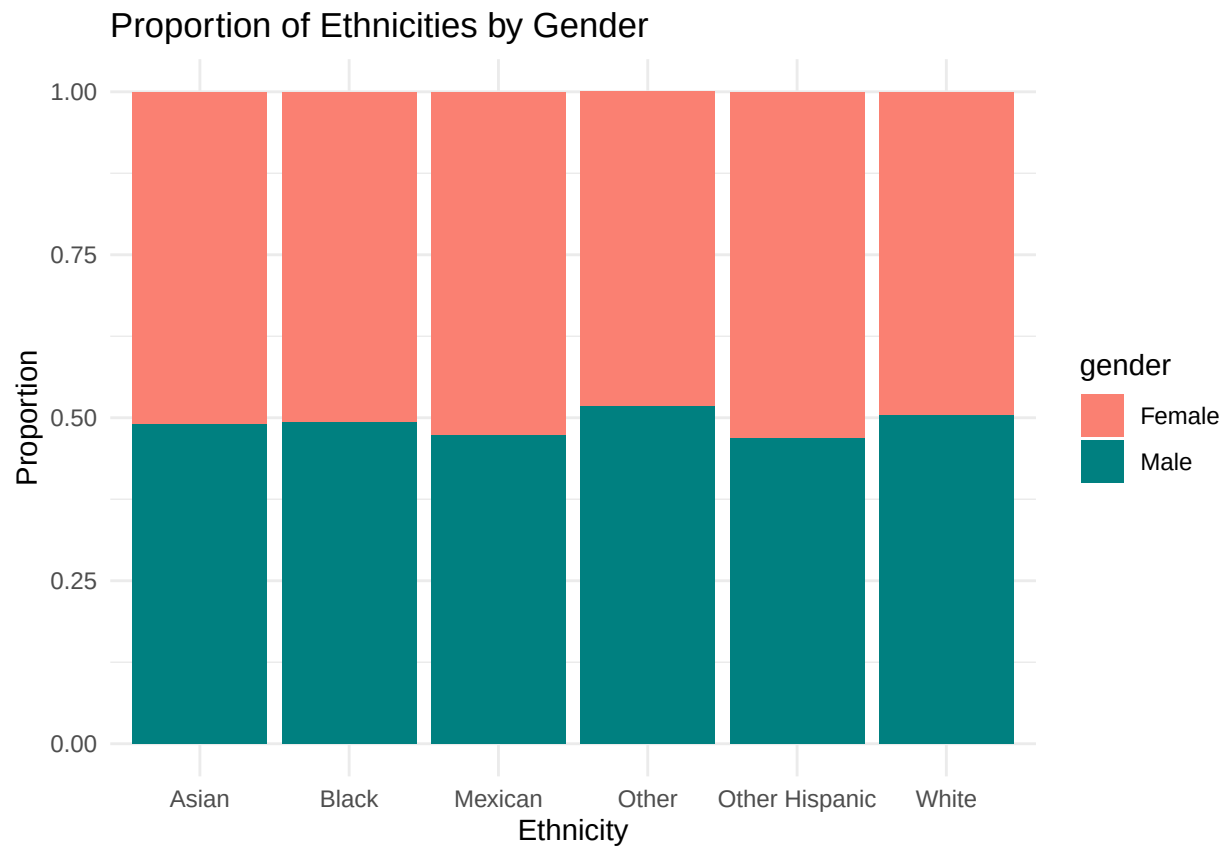
Figure 2: Stacked bar chart of ethnicity by gender (100% stacked)

```
p2 <- nh %>%  
  ggplot(aes(x = ethnicity_2, fill = gender)) +  
  geom_bar(position = "fill") +  
  scale_fill_manual(values = c("salmon", "#008080")) +  
  labs(  
    title = "Proportion of Ethnicities by Gender",  
    x = "Ethnicity",  
    y = "Proportion"  
  ) +  
  theme_minimal()  
  theme(plot.title = element_text(size = 12))
```

```
## <theme> List of 1  
## $ plot.title: <ggplot2::element_text>  
## ..@ family      : NULL  
## ..@ face        : NULL  
## ..@ italic      : chr NA  
## ..@ fontweight  : num NA  
## ..@ fontwidth   : num NA  
## ..@ colour      : NULL  
## ..@ size        : num 12  
## ..@ hjust       : NULL
```

```
## ..@ vjust      : NULL
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : NULL
## ..@ debug      : NULL
## ..@ inherit.blank: logi FALSE
## @ complete: logi FALSE
## @ validate: logi TRUE
```

p2

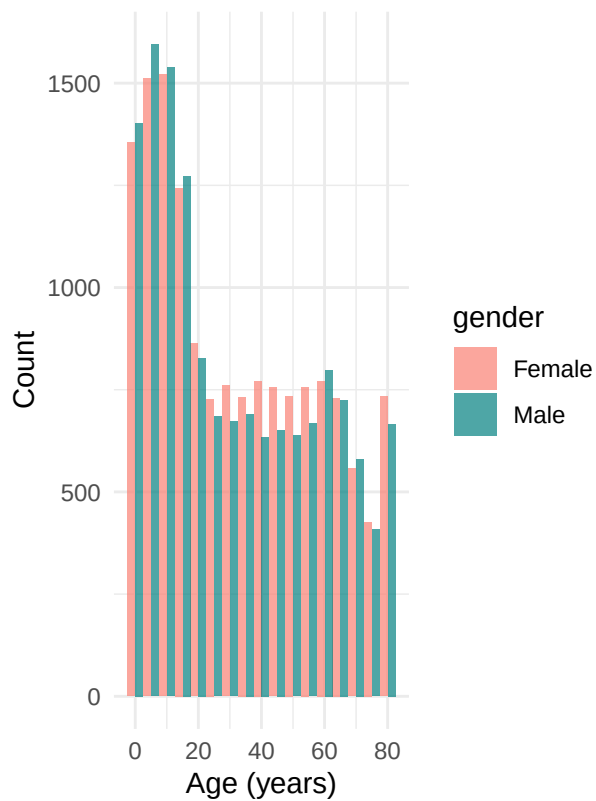


Combined Plots

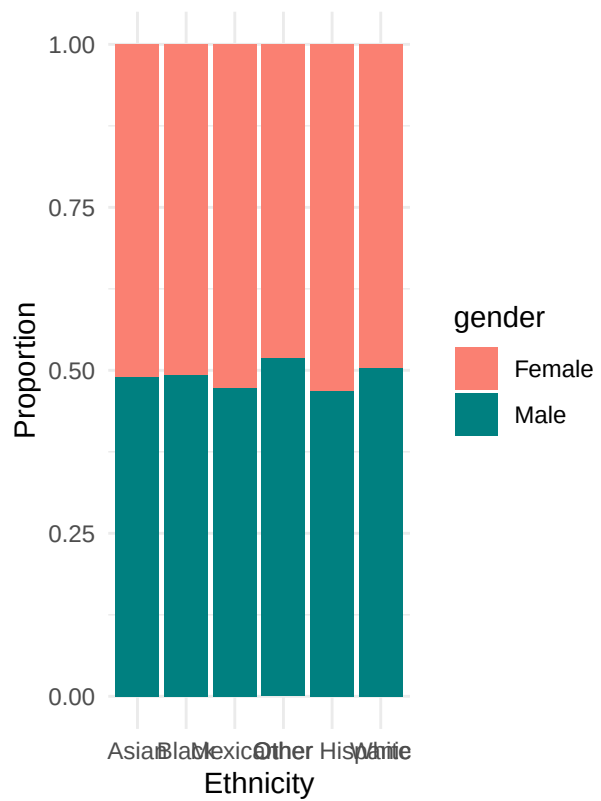
With cowplot

```
combined_cowplot <- plot_grid(p1, p2, labels = c("A", "B"), ncol = 2)
combined_cowplot
```

A Distribution of Age by Gender

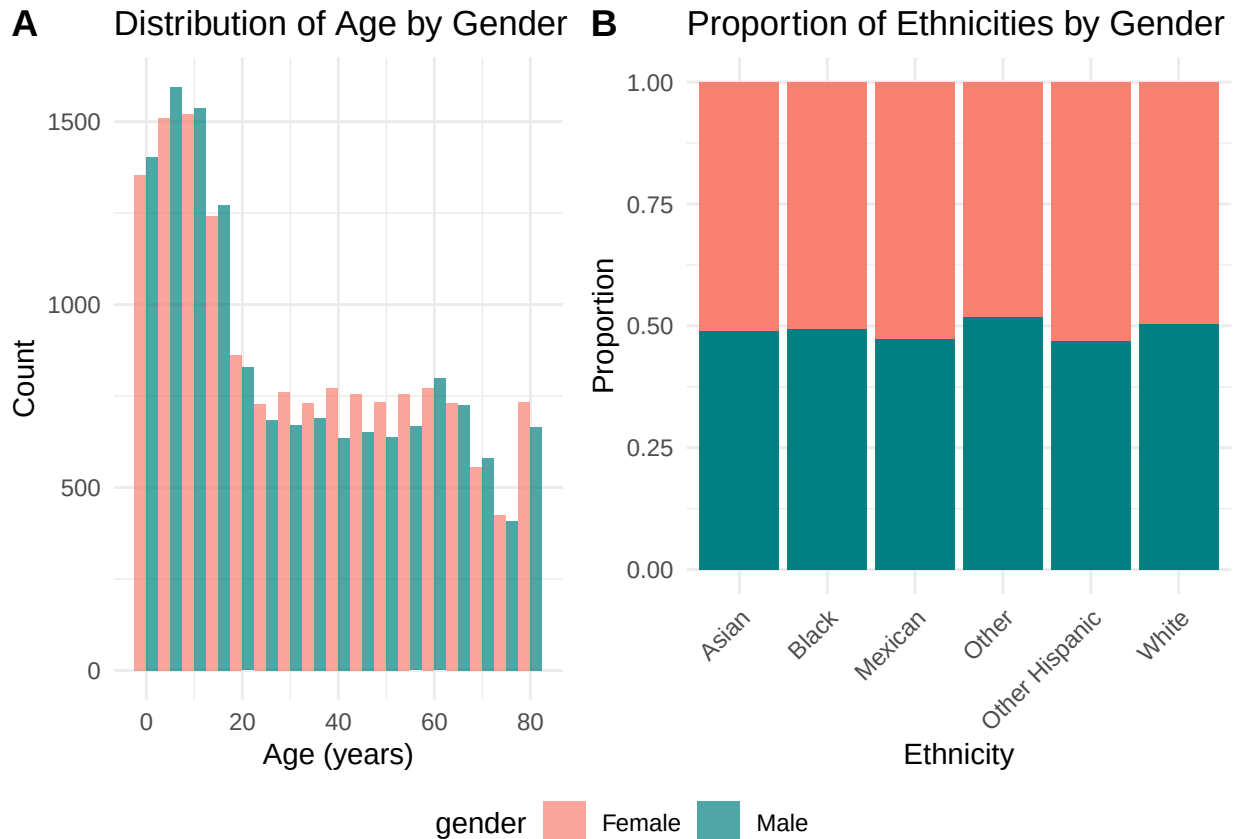


B Proportion of Ethnicities by Gender



```
# With ggpubr
```

```
p2 <- p2 + theme(axis.text.x = element_text(angle = 45, hjust = 1))
combined_ggpubr <- ggarrange(p1, p2, labels = c("A", "B"),
                             ncol = 2, widths = c(1.3, 1.5),
                             common.legend = TRUE, legend = "bottom")
combined_ggpubr
```



Comparing `plot_grid()` and `ggarrange()`

`plot_grid()` (cowplot) is lightweight and gives fine control over axis alignment and the A/B labels. It is excellent when you want perfectly aligned panels.

`ggarrange()` (ggpubr) offers extra features: automatic legend alignment, shared axis labels, and richer layout options.

Based on the core uses of these respective functions, `ggarrange()` has a better layout for both of these plots.

For these two plots, `ggarrange()` generally produces a cleaner result because it aligns the two panels automatically and lets you share a common legend — reducing redundancy since both figures use the same gender colors. To elaborate, `ggarrange()` has “`common.legend = TRUE`”, which neatly gives you a single “gender” legend instead of two. `plot_grid()` doesn’t do this automatically. Additionally, it aligns the panels nicely, allowing for axis alignment to look polished while finalizing the figure.

Exercise 2 — Visualizing key characteristics

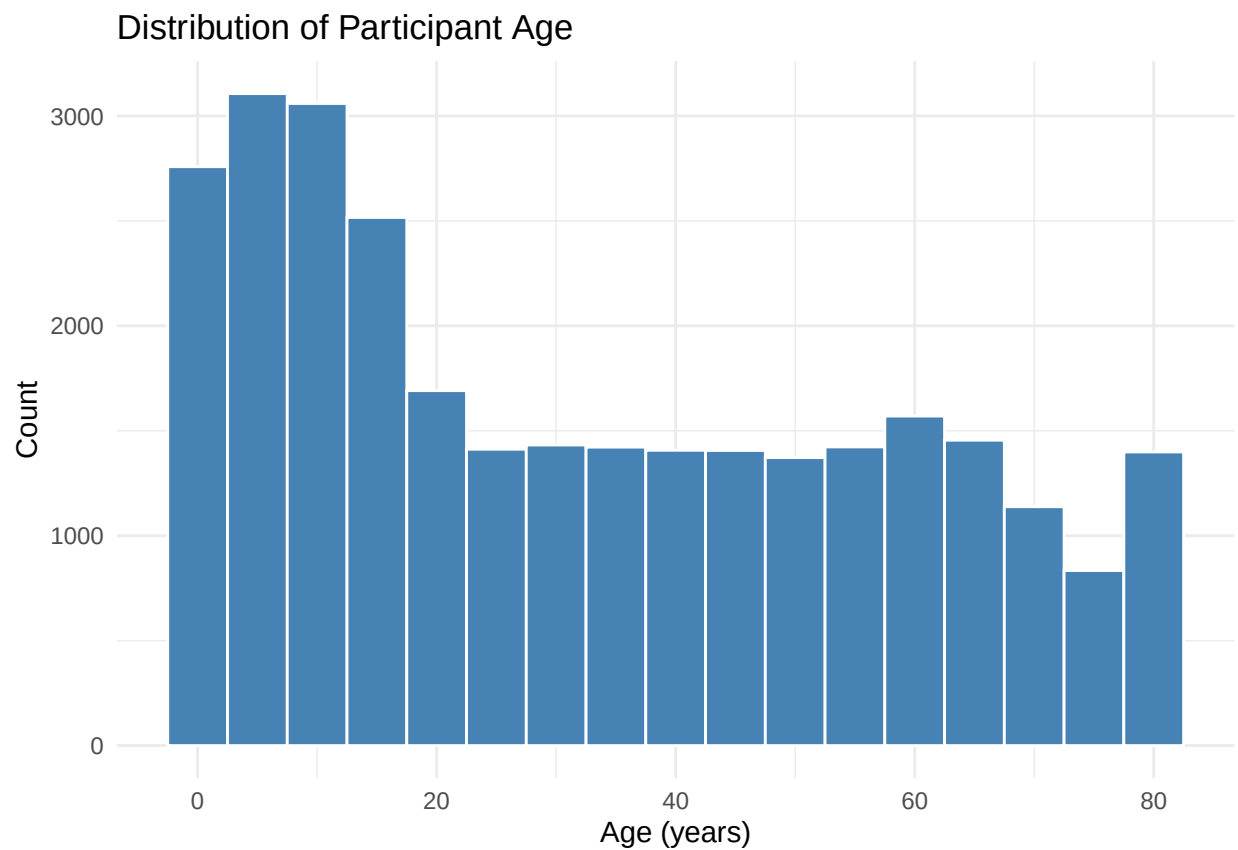
The four plots

```
nh <- read_csv("Data /cleaned_NHANES.csv")
# Adjust column names to match your actual cleaned dataset
```

Age

```
p_age <- nh %>%  
  ggplot(aes(x = age)) +  
  geom_histogram(binwidth = 5, fill = "steelblue", color = "white") +  
  labs(  
    title = "Distribution of Participant Age",  
    x = "Age (years)",  
    y = "Count"  
  ) +  
  theme_minimal()
```

p_age



Gender

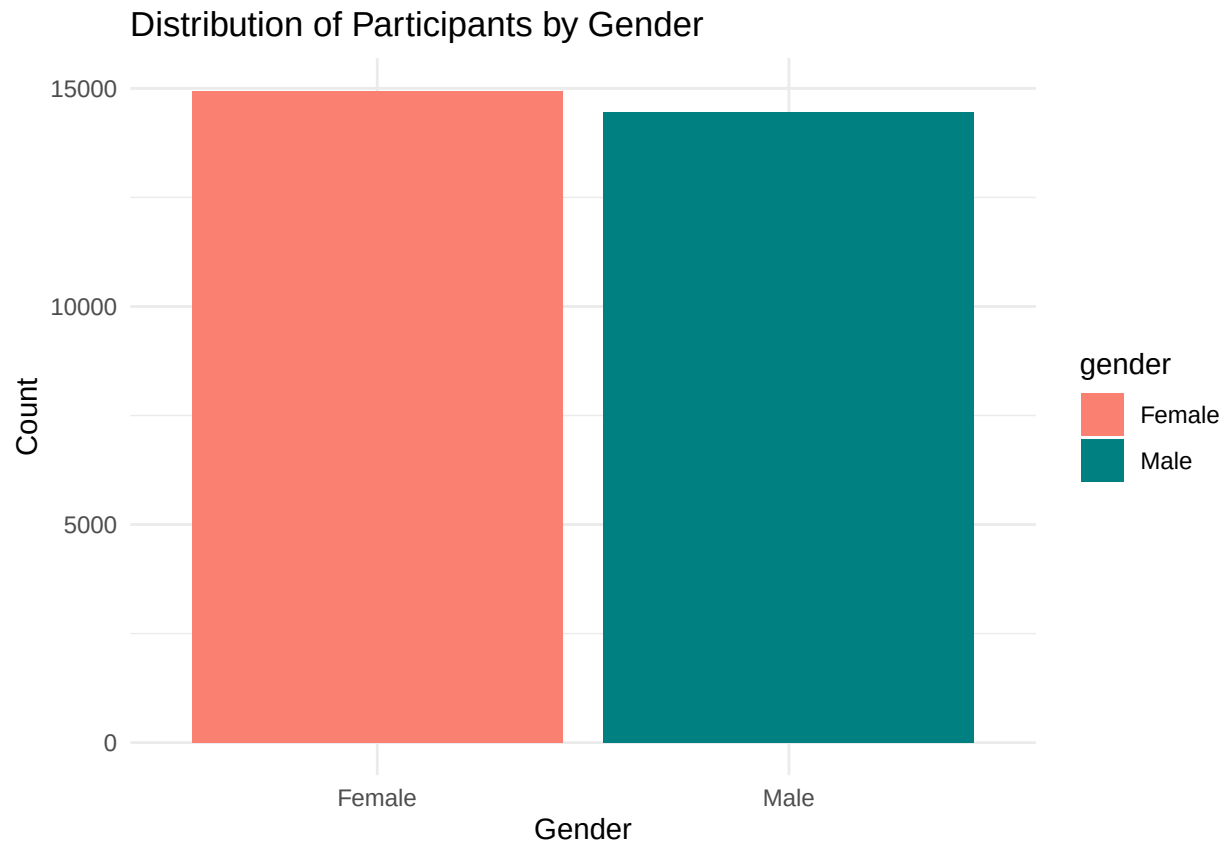
```
p_gender <- nh %>%  
  ggplot(aes(x = gender, fill = gender)) +  
  geom_bar() +  
  scale_fill_manual(values = c("#FA8072", "#008080")) +  
  labs(  
    title = "Distribution of Participant Gender",  
    x = "Gender",  
    y = "Count"  
  ) +  
  theme_minimal()
```

```

    title = "Distribution of Participants by Gender",
    x = "Gender",
    y = "Count"
  ) +
  theme_minimal()

```

p_gender

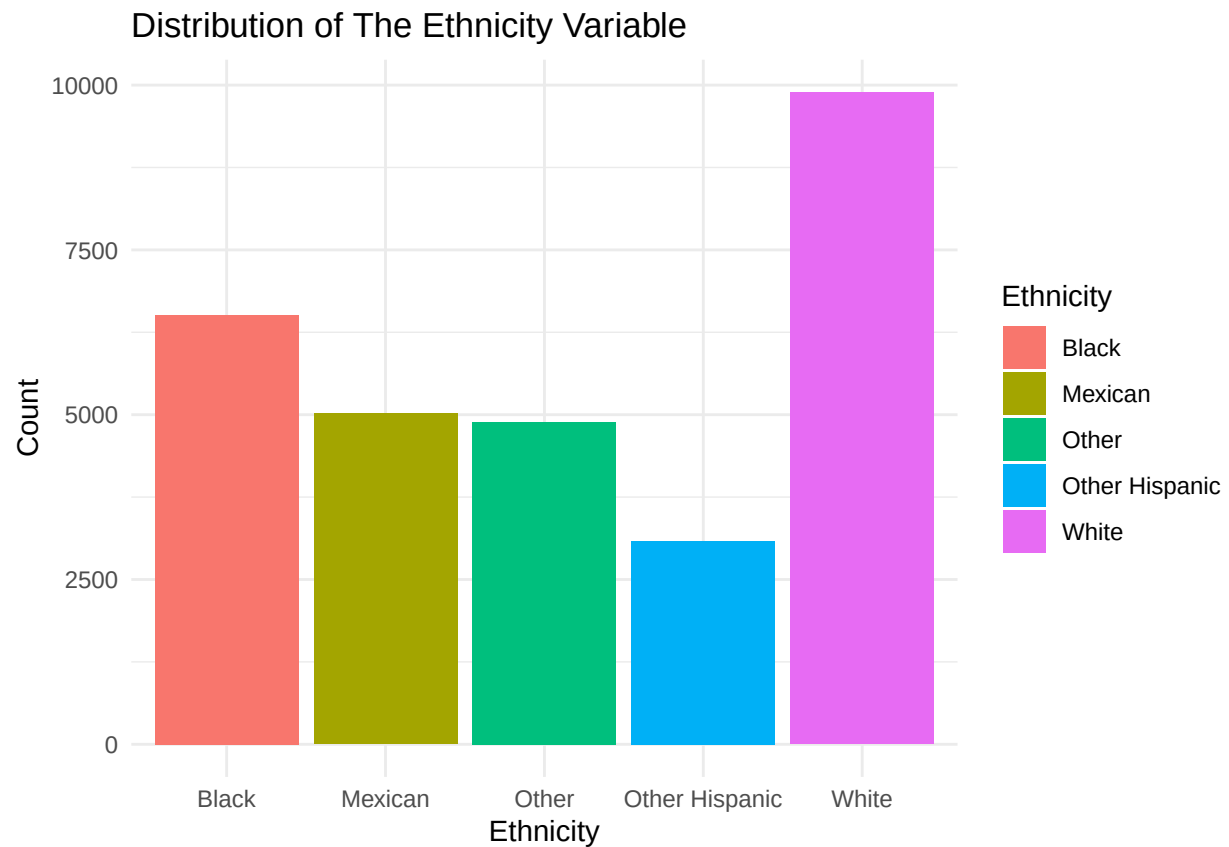


First Ethnicity Variable

```

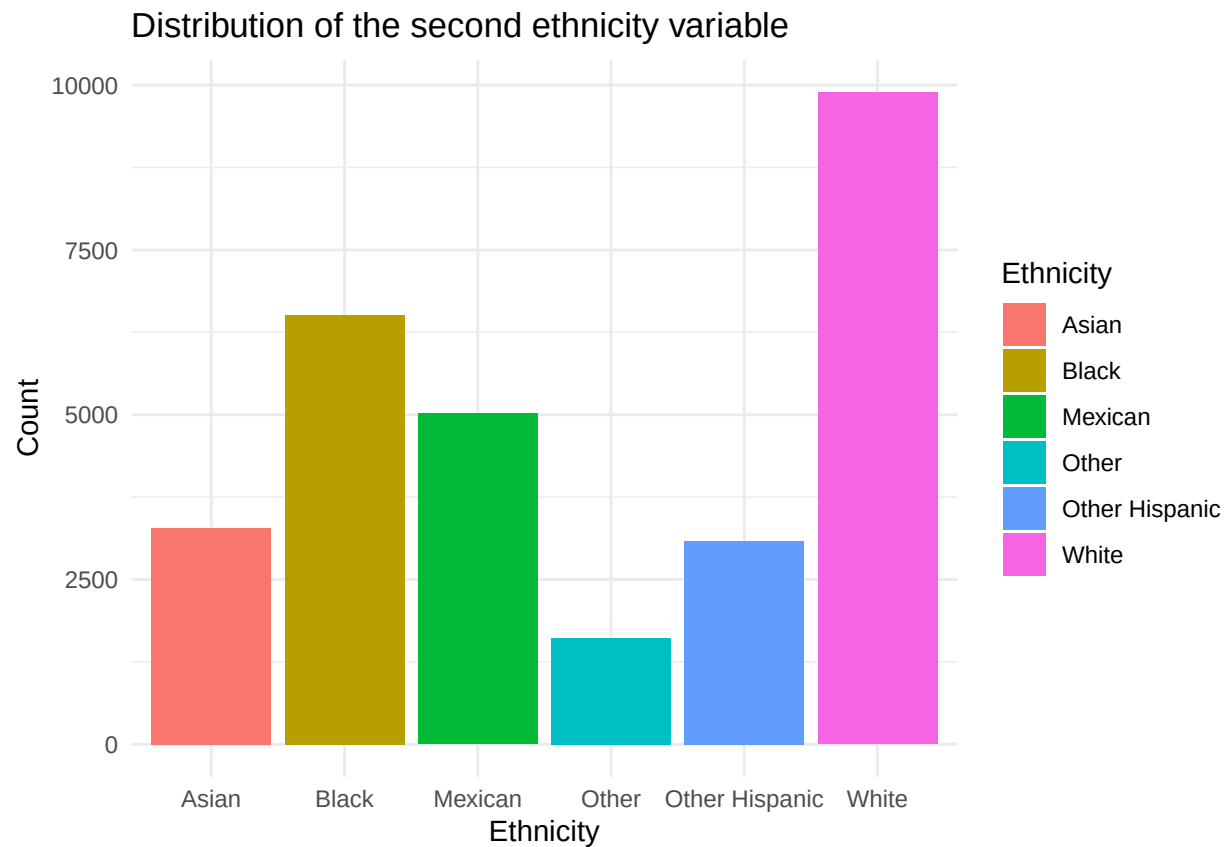
p_eth1 <- nh %>%
  ggplot(aes(x = ethnicity_1 , fill = ethnicity_1)) +
  geom_bar() +
  labs(
    title = "Distribution of The Ethnicity Variable",
    x = "Ethnicity",
    y = "Count"
  ) +
  theme_minimal()
p_eth1 + scale_fill_discrete(name = "Ethnicity")

```



Second Ethnicity Variable

```
p_eth2 <- nh %>%  
  ggplot(aes(x = ethnicity_2, fill = ethnicity_2)) +  
  geom_bar() +  
  labs(  
    title = "Distribution of the second ethnicity variable",  
    x = "Ethnicity",  
    y = "Count"  
  ) +  
  theme_minimal()  
p_eth2 + scale_fill_discrete(name = "Ethnicity")
```



Missing values

```
sum(is.na(nh$age))
```

```
## [1] 0
```

```
sum(is.na(nh$gender))
```

```
## [1] 0
```

```
sum(is.na(nh$ethnicity_1))
```

```
## [1] 0
```

```
sum(is.na(nh$ethnicity_2))
```

```
## [1] 0
```

Describing the distributions

Age: roughly uniform across the adult range (the NHANES sample is designed to be representative), with missing values as reported above.

Gender: near-balanced split between males and females.

Ethnicity: uneven distribution, with some groups much more represented than others. Missing values as reported above.

Saving the plots

```
ggsave("age_plot.png", p_age, width = 8, height = 5)
ggsave("gender_plot.png", p_gender, width = 8, height = 5)
ggsave("ethnicity_1_plot.png", p_eth1, width = 8, height = 5)
ggsave("ethnicity_2_plot.png", p_eth2, width = 8, height = 5)
```

Comparing the two ethnicity variables

```
# Count how many categories each has
length(unique(nh$ethnicity_1))
```

```
## [1] 5
```

```
length(unique(nh$ethnicity_2))
```

```
## [1] 6
```

NHANES contains RIDRETH1 (5 categories) and RIDRETH3 (6 categories, adding other races and multi-racial). Thus, I will keep the second ethnicity sample because it is more detailed and reflects NHANES's current classification and remove the 5-category variable.

```
nh <- nh %>% select(-ethnicity_1)
```

Exercise 3 — Improving ggplot Figure

Problems with the original figure

1. **No baseline reference:** the plot shows absolute weight, but the goal is to compare each participant against their own weight at week 0. The figure should plot the **weight change** relative to week 0 instead.
2. **Visual clutter:** 19 distinct colored lines that overlap heavily and are rather hard to read. Furthermore, the legend is crowded and the lines cross each other.
3. **No zero reference line:** without one, it's difficult to immediately see who lost vs. gained weight.
4. **Generic axis labels:** "Weight" and "Week" lack units and context; weight should be labeled as change in kg, and the week scale should be clear.

The improved figure

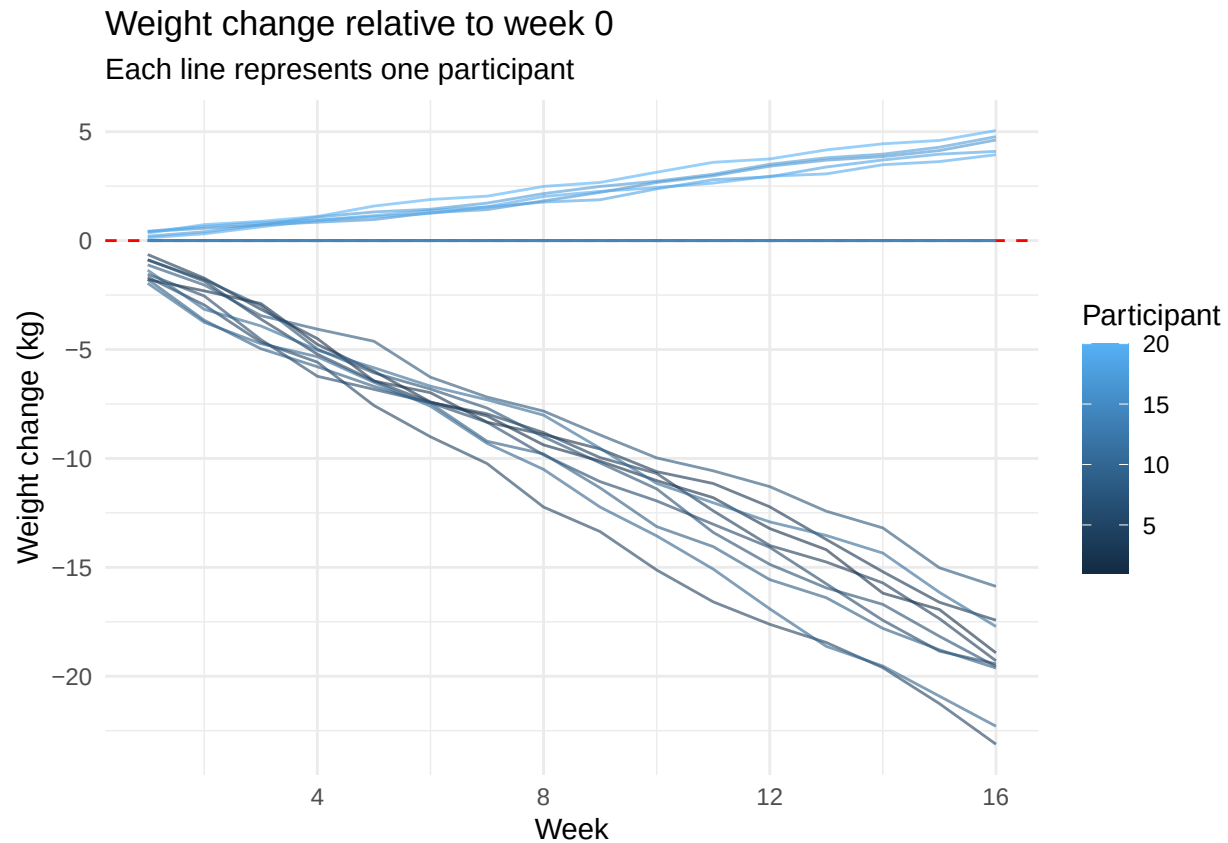
```
diet <- read_csv("Data /diet.csv")

# Compute weight change relative to week 0
diet_wide <- diet %>%
  pivot_wider(names_from = Week, values_from = Weight, names_prefix = "w")

diet_change <- diet_wide %>%
  mutate(across(w0:w16, ~ . - w0)) %>% # change relative to week 0
  select(-w0) %>%
  pivot_longer(-Participant, names_to = "Week", values_to = "change") %>%
  mutate(Week = as.numeric(gsub("w", "", Week)))

p_diet <- diet_change %>%
  ggplot(aes(x = Week, y = change, group = Participant, color = Participant)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  geom_line(alpha = 0.6) +
  labs(
    title = "Weight change relative to week 0",
    subtitle = "Each line represents one participant",
    x = "Week",
    y = "Weight change (kg)",
    color = "Participant"
  ) +
  theme_minimal()

p_diet
```



Why these changes help

- **Relative change** directly shows the diet's effect, which is the actual goal of the study. A participant who lost 3 kg shows a line at -3 , regardless of their starting weight.
- **The zero line** makes loss (below 0) vs. gain (above 0) instantly readable.
- **Reduced opacity** ($\alpha = 0.6$) keeps overlapping lines distinguishable despite 19 participants.

Exercise 4— Exploring relationships through visualizations

Step 1: Average SBP variable

```
nh <- nh %>%
  mutate(
    avg_sbp = rowMeans(select(., systolic_bp_1, systolic_bp_2, systolic_bp_3, systolic_bp_4), na.rm = TRUE),
    avg_sbp = ifelse(
      is.na(systolic_bp_1) & is.na(systolic_bp_2) & is.na(systolic_bp_3) & is.na(systolic_bp_4),
      NA, avg_sbp
    )
  )
```

Step 2: Filter to non-missing values

```
nh_final <- nh %>%  
  filter(!is.na(age), !is.na(gender), !is.na(avg_sbp))
```

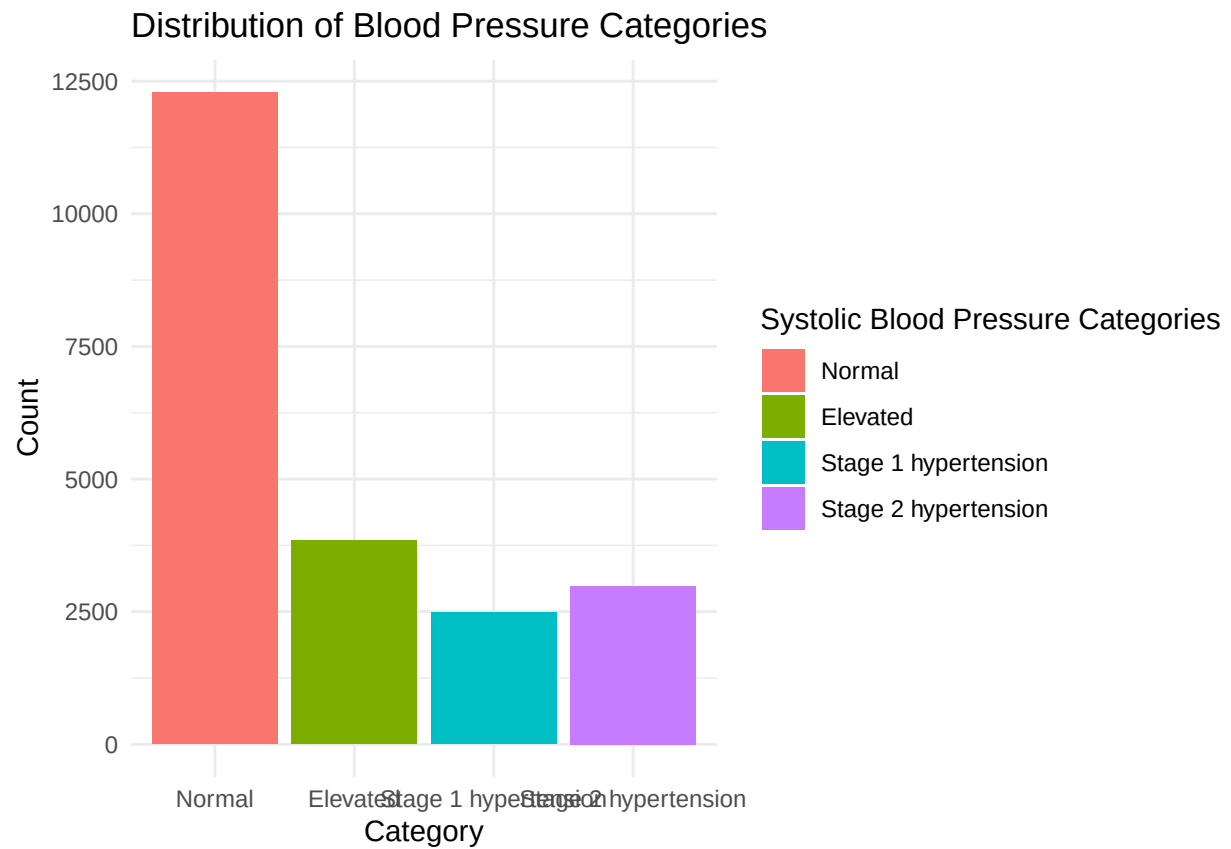
Step 3: Categorize blood pressure

```
nh_final <- nh_final %>%  
  mutate(  
    sbp_cat = case_when(  
      avg_sbp < 120 ~ "Normal",  
      avg_sbp >= 120 & avg_sbp < 130 ~ "Elevated",  
      avg_sbp >= 130 & avg_sbp < 140 ~ "Stage 1 hypertension",  
      avg_sbp >= 140 ~ "Stage 2 hypertension"  
    ),  
    sbp_cat = factor(sbp_cat, levels = c(  
      "Normal", "Elevated", "Stage 1 hypertension", "Stage 2 hypertension"  
    ))  
  )
```

Step 4: Up to four plots describing the final sample

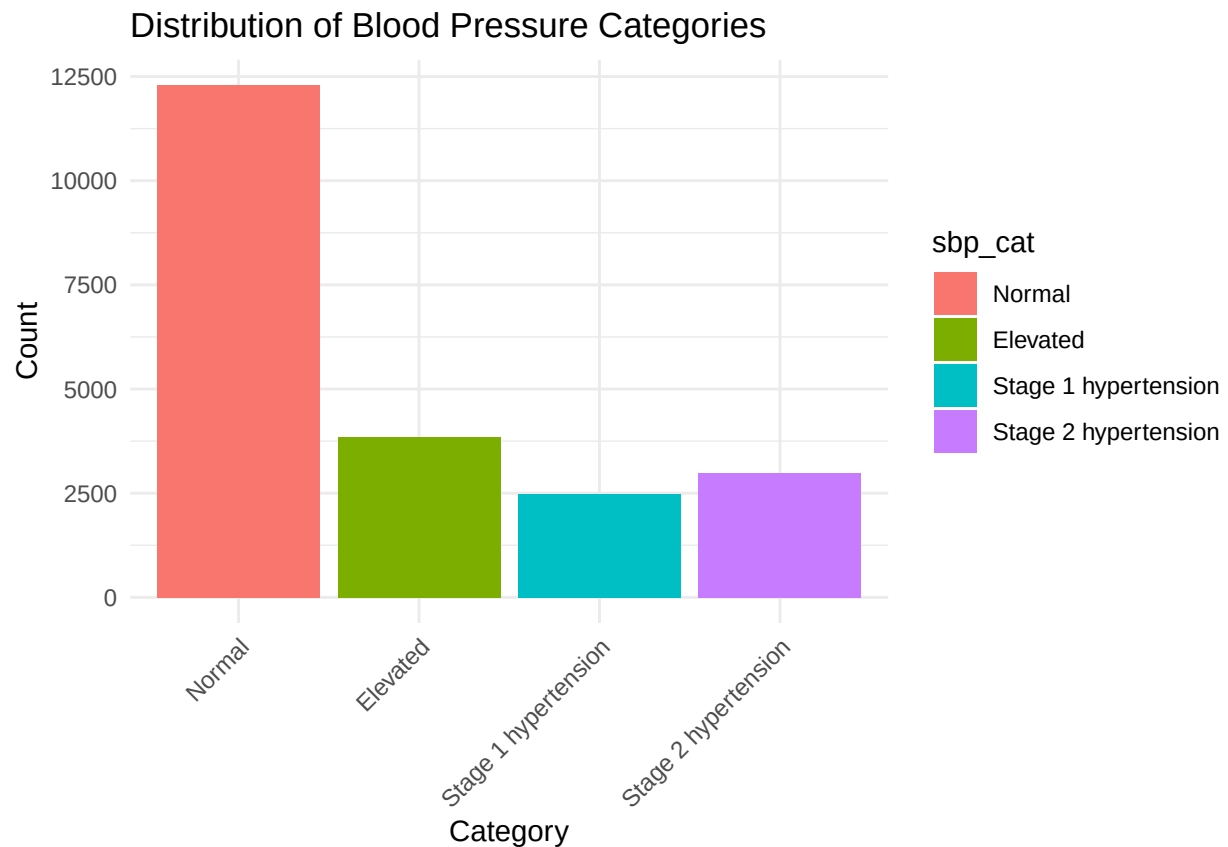
1. Blood pressure category counts

```
p_cat <- nh_final %>%  
  ggplot(aes(x = sbp_cat, fill = sbp_cat)) +  
  geom_bar() +  
  labs(title = "Distribution of Blood Pressure Categories",  
        x = "Category", y = "Count") +  
  theme_minimal()  
p_cat + scale_fill_discrete(name = "Systolic Blood Pressure Categories")
```



```
p_cat <- p_cat + theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

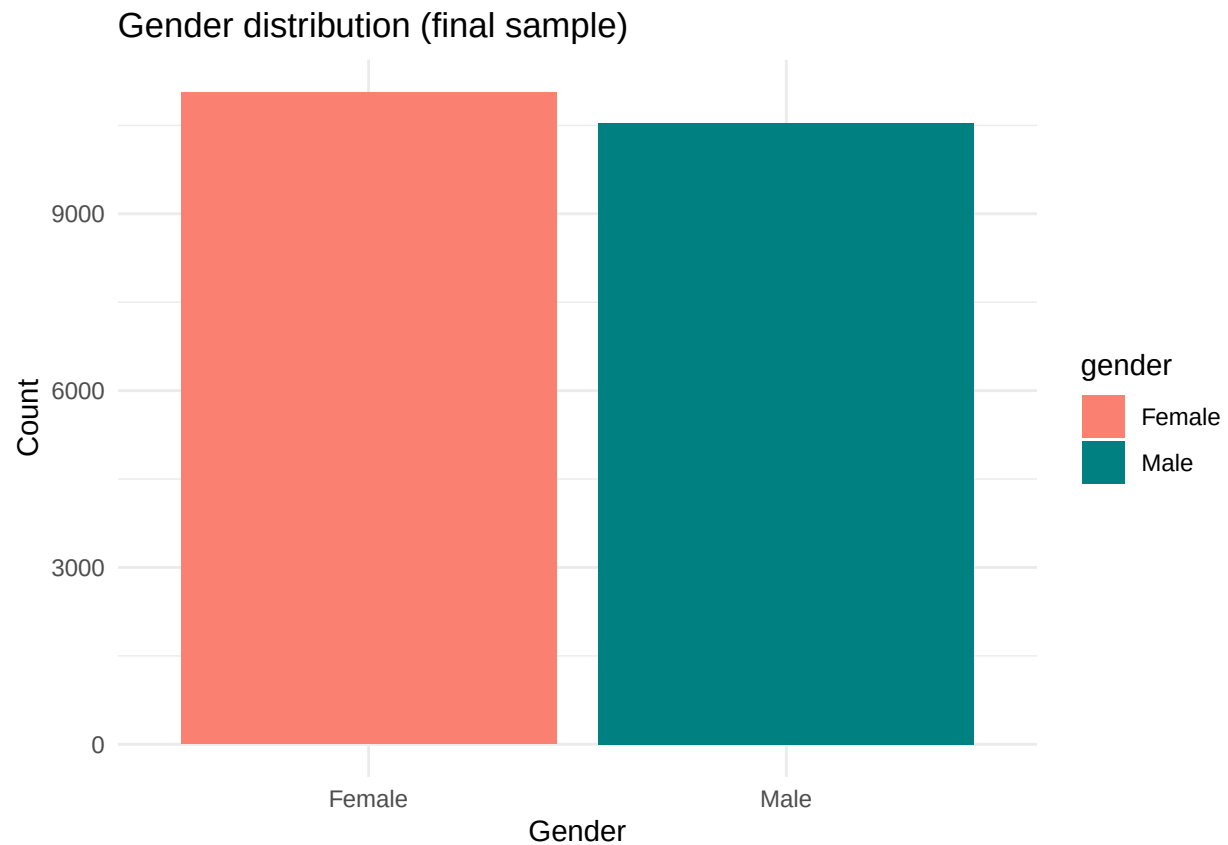
```
p_cat
```



2. Gender distribution in the final sample

```
p_gender_final <- nh_final %>%
  ggplot(aes(x = gender, fill = gender)) +
  geom_bar() +
  scale_fill_manual(values = c("#FA8072", "#008080")) +
  labs(title = "Gender distribution (final sample)",
       x = "Gender", y = "Count") +
  theme_minimal()

p_gender_final
```



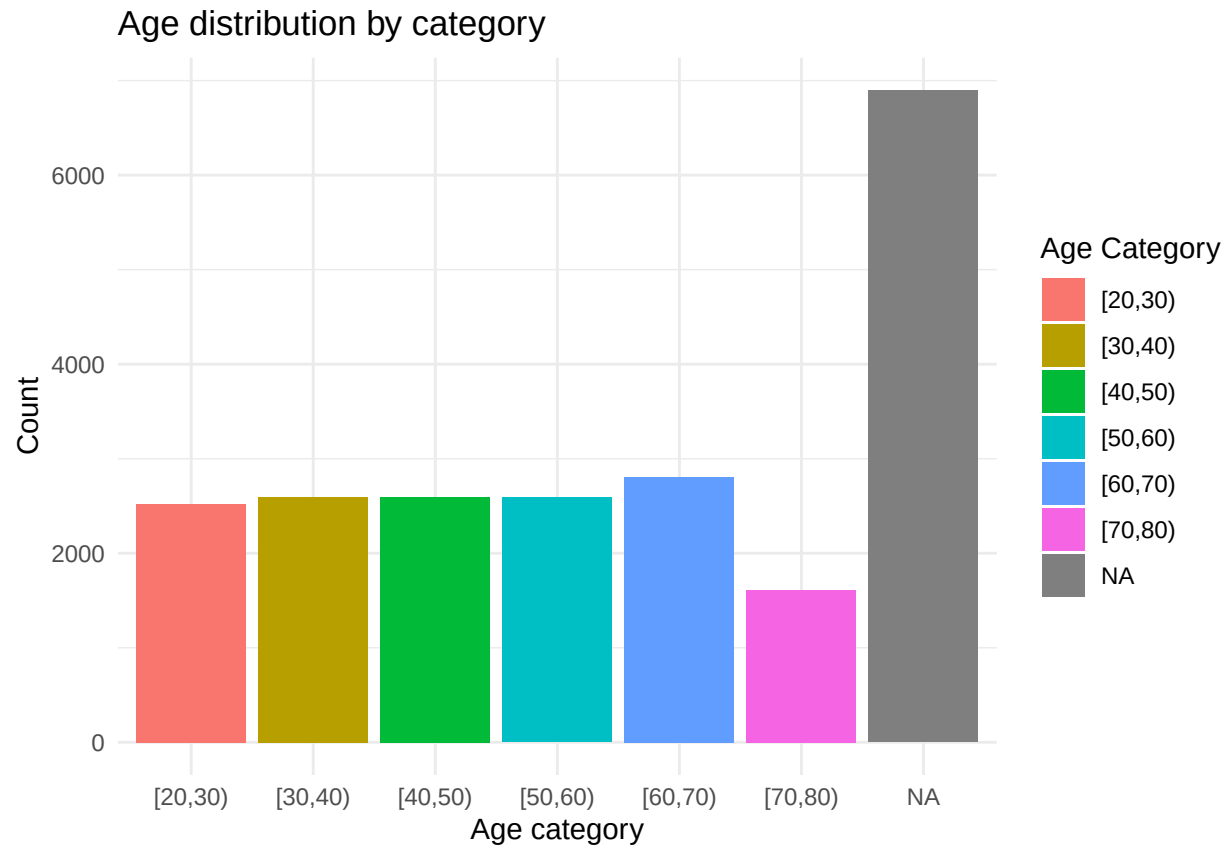
3. Reuse the age plot saved in Exercise 2

```
# Reuse the saved "age_plot.png" from Exercise 2
```

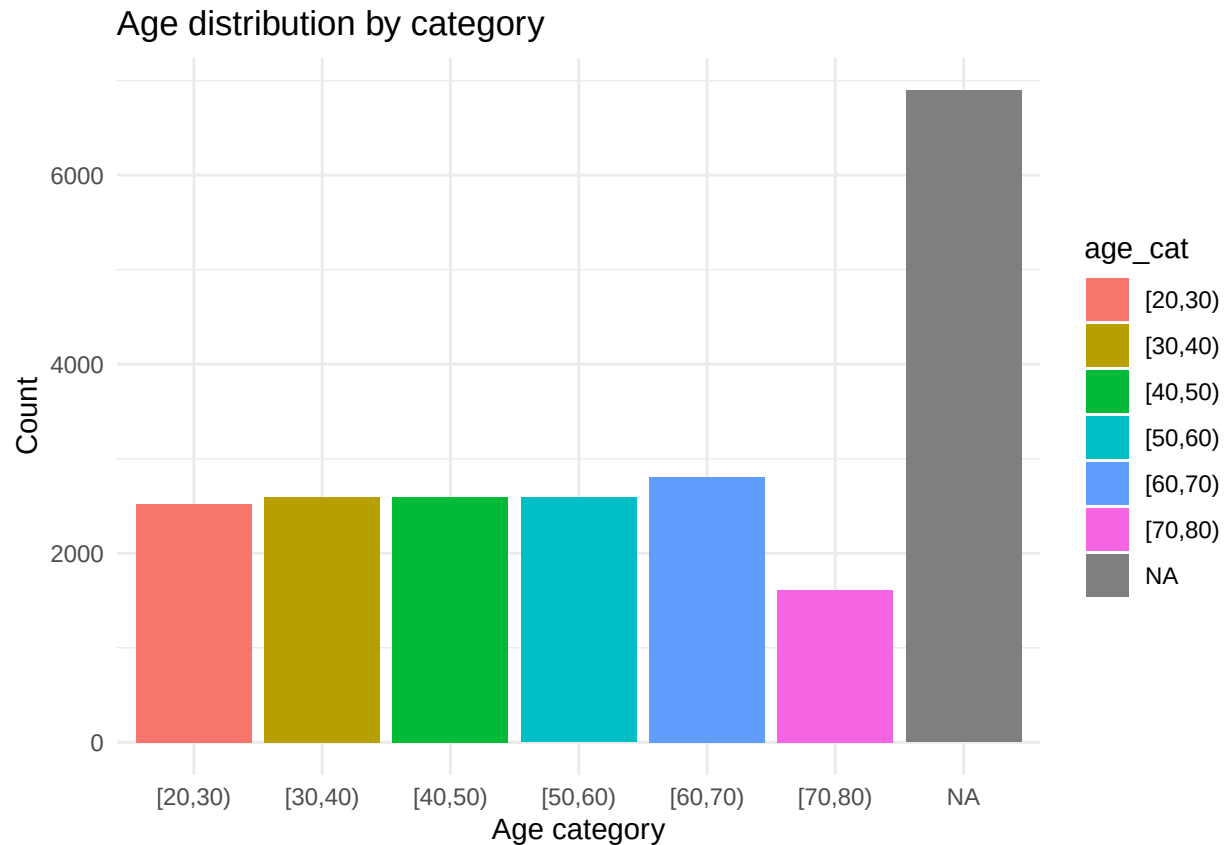
4. Categorized age

```
nh_final <- nh_final %>%
  mutate(age_cat = cut(age, breaks = c(20, 30, 40, 50, 60, 70, 80),
    right = FALSE))

p_agecat <- nh_final %>%
  ggplot(aes(x = age_cat, fill = age_cat)) +
  geom_bar() +
  labs(title = "Age distribution by category",
    x = "Age category", y = "Count") +
  theme_minimal()
p_agecat + scale_fill_discrete(name = "Age Category")
```



p_agecat



Sample size change

```
nrow(nh)           # before filtering
```

```
## [1] 29400
```

```
nrow(nh_final)     # after filtering
```

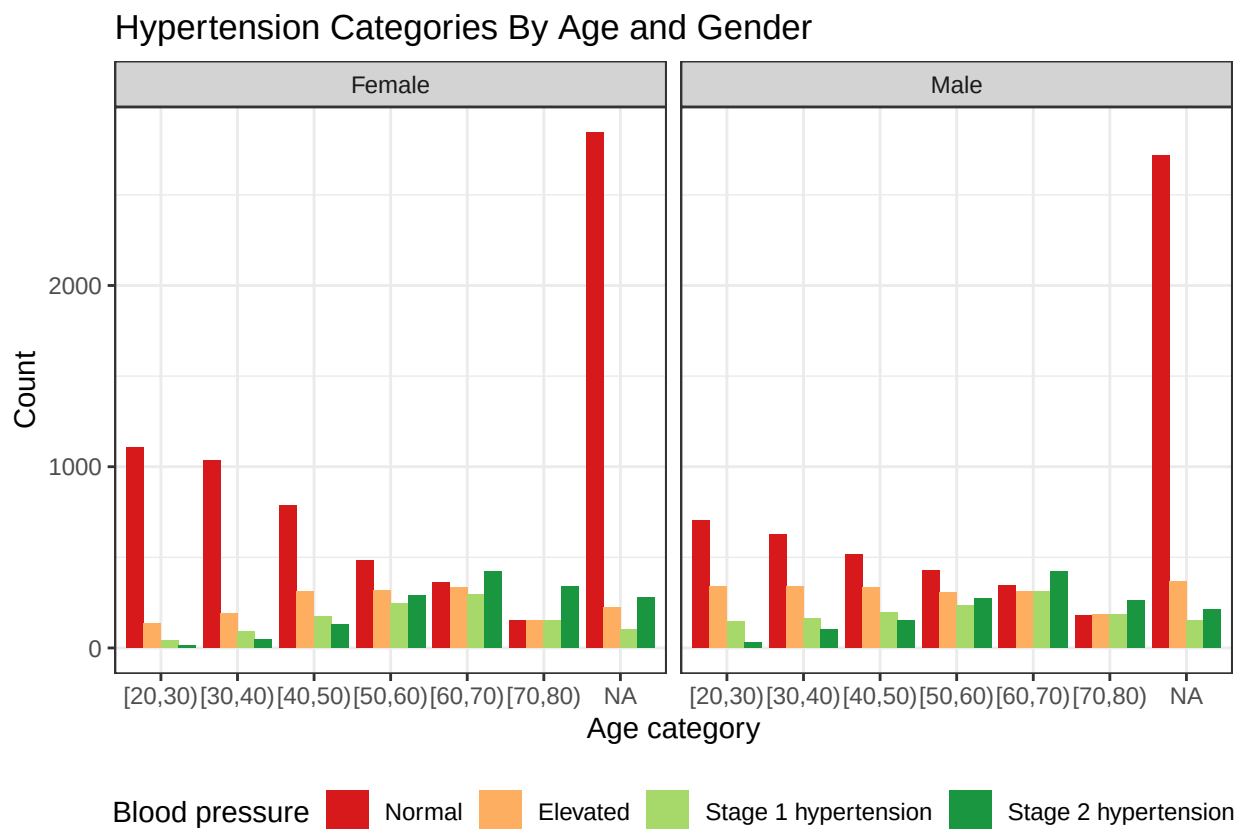
```
## [1] 21604
```

After removing individuals with missing blood pressure readings, the final sample is smaller than the full dataset. This is expected as the reduction mainly affects younger adults, who are less likely to complete the physical exam component that provides the SBP readings. As a result, the age distribution of the final sample skews slightly older than the full sample.

Step 5: Hypertension categories across age and gender

```
p_hyp <- nh_final %>%
  ggplot(aes(x = age_cat, fill = sbp_cat)) +
  geom_bar(position = "dodge") +
  facet_wrap(~ gender) +
  scale_fill_brewer(palette = "RdYlGn") +
  labs(
    title = "Hypertension Categories By Age and Gender",
    x = "Age category",
    y = "Count",
    fill = "Blood pressure"
  ) +
  theme_bw() +
  theme(strip.background = element_rect(fill = "lightgrey"),
        legend.position = "bottom")

p_hyp
```



Report

This report explores how age and gender relate to systolic blood pressure (SBP) in the NHANES sample. After computing an average SBP from the four readings and removing individuals with missing values, the final sample retained 21,604 participants. The sample is roughly balanced by gender, with an age distribution that skews slightly older than the full dataset, since blood pressure readings are more likely to be available for adults who completed the exam component.

The results show that blood pressure categories shift systematically with age. Normal blood pressure is most common among younger adults, while the proportion of elevated, stage 1, and stage 2 hypertension rises steadily with age. This pattern appears for both genders, though the faceted plot reveals that men tend to reach higher blood pressure categories at younger ages than women.